

Appendix A - Code Order

Full PACP Code	Description	Distance (Feet)	Video Ref.	Code		Continuous	Value				Joint	Location		Image Ref.	Remarks
				Group/Descriptor	Modifier/Severity		S/M/L	Inches		%		At/ From	To		
								1 st	2 nd						
ACB	Catch Basin	Required	Optional	ACB									Optional	Optional	
ACOH	Cleanout House	Required	Optional	ACO	H								Optional	Optional	
ACOM	Cleanout Mainline	Required	Optional	ACO	M								Optional	Required	
ACOP	Cleanout Propertyline	Required	Optional	ACO	P								Optional	Optional	
ADP	Discharge Point	Required	Optional	ADP									Optional	Optional	
AEP	End of Pipe	Required	Optional	AEP									Optional	Optional	
AJB	Junction Box	Required	Optional	AJB									Optional	Required	
AM	Meter	Required	Optional	AM									Optional	Required	
AMH	Manhole	Required	Optional	AMH									Optional	Required	
AOC	Special Chamber	Required	Optional	AOC									Optional	Optional	
ATC	Tee Connection	Required	Optional	ATC			Required	Optional			Required		Optional	Optional	
AWA	Wastewater Access Device	Required	Optional	AWA									Optional	Required	
AWW	Wet Well	Required	Optional	AWW									Optional	Optional	
B	Broken	Required	Optional	B		Optional	Optional	Optional		Optional	Required	Optional	Optional	Optional	
BSV	Broken Soil Visible	Required	Optional	B	SV	Optional	Optional	Optional		Optional	Required	Optional	Optional	Optional	
BVV	Broken Void Visible	Required	Optional	B	VV	Optional	Optional	Optional		Optional	Required	Optional	Optional	Optional	
CC	Crack Circumferential	Required	Optional	CC		Optional	Optional	Optional		Optional	Required	Required	Optional	Optional	
CH2	Crack Longitudinal Hinge, 2	Required	Optional	CH2		Optional	Optional	Optional		Optional	Required	Required	Optional	Optional	
CH3	Crack Longitudinal Hinge, 3	Required	Optional	CH3		Optional	Optional	Optional		Optional	Required	Required	Optional	Optional	
CH4	Crack Longitudinal Hinge, 4	Required	Optional	CH4		Optional	Optional	Optional		Optional	Required	Required	Optional	Optional	
CL	Crack Longitudinal	Required	Optional	CL		Optional	Optional	Optional		Optional	Required		Optional	Optional	
CM	Crack Multiple	Required	Optional	CM		Optional	Optional	Optional		Optional	Required	Required	Optional	Optional	
CS	Crack Spiral	Required	Optional	CS		Optional	Optional	Optional		Optional	Required	Required	Optional	Optional	
D	Deformed	Required	Optional	D		Optional		Required 5% increments					Optional	Optional	
DAE	Deposits Attached Encrustation	Required	Optional	DA	E	Optional		Required		Optional	Required	Optional	Optional	Optional	
DAGS	Deposits Attached Grease	Required	Optional	DA	GS	Optional		Required		Optional	Required	Optional	Optional	Optional	
DAR	Deposits Attached Ragging	Required	Optional	DA	R	Optional		Required		Optional	Required	Optional	Optional	Optional	
DAZ	Deposits Attached Other	Required	Optional	DA	Z	Optional		Required		Optional	Required	Optional	Optional	Optional	
DB	Displaced Brick	Required	Optional	DB		Optional					Required	Optional	Optional	Optional	
DH	Deformed Horizontal	Required	Optional	DH		Optional		Required 5% increments					Optional	Optional	
DI	Dropped Invert	Required	Optional	DI		Optional	Required						Optional	Optional	
DNF	Deposits Ingressed Fine	Required	Optional	DN	F	Optional		Required		Optional	Required	Optional	Optional	Optional	
DNGV	Deposits Ingressed Gravel	Required	Optional	DN	GV	Optional		Required		Optional	Required	Optional	Optional	Optional	
DNZ	Deposits Ingressed Other	Required	Optional	DN	Z	Optional		Required		Optional	Required	Optional	Optional	Required	
DSC	Deposits Settled Compacted	Required	Optional	DS	C	Optional		Required		Optional	Required	Optional	Optional	Optional	
DSF	Deposits Settled Fine	Required	Optional	DS	F	Optional		Required		Optional	Required	Optional	Optional	Optional	
DSGV	Deposits Settled Gravel	Required	Optional	DS	GV	Optional		Required		Optional	Required	Optional	Optional	Optional	
DSZ	Deposits Settled Other	Required	Optional	DS	Z	Optional		Required		Optional	Required	Optional	Optional	Required	
DV	Deformed Vertical	Required	Optional	DV		Optional		Required 5% increments					Optional	Optional	
FC	Fracture Circumferential	Required	Optional	FC		Optional	Optional	Optional		Optional	Required	Required	Optional	Optional	
FH2	Fracture Longitudinal Hinge, 2	Required	Optional	FH2		Optional	Optional	Optional		Optional	Required	Required	Optional	Optional	
FH3	Fracture Longitudinal Hinge, 3	Required	Optional	FH3		Optional	Optional	Optional		Optional	Required	Required	Optional	Optional	
FH4	Fracture Longitudinal Hinge, 4	Required	Optional	FH4		Optional	Optional	Optional		Optional	Required	Required	Optional	Optional	
FL	Fracture Longitudinal	Required	Optional	FL		Optional	Optional	Optional		Optional	Required		Optional	Optional	
FM	Fracture Multiple	Required	Optional	FM		Optional	Optional	Optional		Optional	Required	Required	Optional	Optional	
FS	Fracture Spiral	Required	Optional	FS		Optional	Optional	Optional		Optional	Required	Required	Optional	Optional	

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				Group/Descriptor	Modifier/Severity		S/M/L	Inches		%		At/ From	To		
								1 st	2 nd						
GRT	Grout done at Location	Required	Optional					Optional						Optional	Optional
GTFJ	Grout Air Test Fail Joint	Required	Optional					Optional	Required					Optional	Optional
GTFL	Grout Air Test Fail Lateral	Required	Optional					Optional	Required					Optional	Optional
GTPJ	Grout Air Test Pass Joint	Required	Optional											Optional	Optional
GTPL	Grout Air Test Pass Lateral	Required	Optional											Optional	Optional
GTUJ	Grout Air Test Unable Joint	Required	Optional											Optional	Required
GTUL	Grout Air Test Unable Lateral	Required	Optional											Optional	Required
H	Hole	Required	Optional	H		Optional		Optional	Optional		Optional	Required	Optional	Optional	Optional
HSV	Hole Soil Visible	Required	Optional	H	SV	Optional		Optional	Optional		Optional	Required	Optional	Optional	Optional
HVV	Hole Void Visible	Required	Optional	H	VV	Optional		Optional	Optional		Optional	Required	Optional	Optional	Optional
ID	Infil Dripper	Required	Optional	ID		Optional					Optional	Required	Optional	Optional	Optional
IG	Infil Gusher	Required	Optional	IG		Optional					Optional	Required	Optional	Optional	Optional
IR	Infil Runner	Required	Optional	IR		Optional					Optional	Required	Optional	Optional	Optional
IS	Infil Stain	Required	Optional	IS		Optional					Optional	Required	Optional	Optional	Optional
ISGT	Intruding Sealing Grout	Required	Optional	ISGT		Optional				Required		Required	Required	Optional	Optional
ISSR	Intruding Sealing Ring	Required	Optional	ISSR		Optional				Required		Required	Required	Optional	Optional
ISSRB	Intruding Sealing Ring Broken	Required	Optional	ISSR	B	Optional				Required		Required	Required	Optional	Optional
ISSRH	Intruding Sealing Ring Hanging	Required	Optional	ISSR	H	Optional				Required		Required	Required	Optional	Optional
ISSRL	Intruding Sealing Ring Loose/Poorly Fitting	Required	Optional	ISSR	L	Optional				Required		Required	Required	Optional	Optional
ISZ	Intruding Sealing Other	Required	Optional	ISZ		Optional				Required		Required	Required	Optional	Optional
IW	Infil Weeper	Required	Optional	IW		Optional					Optional	Required	Optional	Optional	Optional
JAL	Joint Angular Large	Required	Optional	JA		Optional	L	Optional	Optional					Optional	Optional
JAM	Joint Angular Medium	Required	Optional	JA		Optional	M	Optional	Optional					Optional	Optional
JOL	Joint Offset Large	Required	Optional	JO		Optional	L	Optional	Optional					Optional	Optional
JOM	Joint Offset Medium	Required	Optional	JO		Optional	M	Optional	Optional					Optional	Optional
JSL	Joint Separated Large	Required	Optional	JS		Optional	L	Optional	Optional					Optional	Optional
JSM	Joint Separated Medium	Required	Optional	JS		Optional	M	Optional	Optional					Optional	Optional
KW	Buckling Wall	Required	Optional	K	W	Optional						Required	Optional	Optional	Optional
KD	Buckling Dimpling	Required	Optional	K	D	Optional						Required	Optional	Optional	Optional
KI	Inverse Curvature	Required	Optional	K	I	Optional				required		Required		Optional	Optional
LD	Alignment Down	Required	Optional	LD		Optional				Required				Optional	Optional
LFAC	Lining Failure Abandoned Connection	Required	Optional	LF	AC	Optional					Optional	Required	Optional	Optional	Optional
LFAS	Lining Failure Annular Space	Required	Optional	LF	AS	Optional					Optional	Required	Optional	Optional	Optional
LFB	Lining Failure Blistered	Required	Optional	LF	B	Optional					Optional	Required	Optional	Optional	Optional
LFBK	Lining Failure Buckled	Required	Optional	LF	BK	Optional					Optional	Required	Optional	Optional	Optional
LFBU	Lining Failure Bulges	Required	Optional	LF	BU	Optional					Optional	Required	Optional	Optional	Optional
LFCS	Lining Failure Connection Cut Shifted	Required	Optional	LF	CS	Optional					Optional	Required	Optional	Optional	Optional
LFD	Lining Failure Detached	Required	Optional	LF	D	Optional					Optional	Required	Optional	Optional	Optional
LFDC	Lining Failure Discoloration	Required	Optional	LF	DC	Optional					Optional	Required	Optional	Optional	Optional
LFDE	Lining Failure Defective End	Required	Optional	LF	DE	Optional					Optional	Required	Optional	Optional	Optional
LFDL	Lining Failure Delaminating	Required	Optional	LF	DL	Optional					Optional	Required	Optional	Optional	Optional
LFOC	Lining Failure Overcut Connection	Required	Optional	LF	OC	Optional					Optional	Required	Optional	Optional	Optional
LFPH	Lining Failure Pinhole	Required	Optional	LF	PH	Optional					Optional	Required	Optional	Optional	Optional
LFRS	Lining Failure Resin slug	Required	Optional	LF	RS	Optional					Optional	Required	Optional	Optional	Optional
LFUC	Lining Failure Undercut Connection	Required	Optional	LF	UC	Optional					Optional	Required	Optional	Optional	Optional
LFW	Lining Failure Wrinkled	Required	Optional	LF	W	Optional					Optional	Required	Optional	Optional	Optional
LFZ	Lining Failure Other	Required	Optional	LF	Z	Optional					Optional	Required	Optional	Optional	Required
LL	Alignment Left	Required	Optional	LL		Optional				Required				Optional	Optional

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				Group/Descriptor	Modifier/Severity		S/M/L	Inches		%		At/ From	To		
								1 st	2 nd						
LLD	Alignment Left Down	Required	Optional	LLD		Optional				Required				Optional	Optional
LLU	Alignment Left Up	Required	Optional	LLU		Optional				Required				Optional	Optional
LR	Alignment Right	Required	Optional	LR		Optional				Required				Optional	Optional
LRD	Alignment Right Down	Required	Optional	LRD		Optional				Required				Optional	Optional
LRU	Alignment Right Up	Required	Optional	LRU		Optional				Required				Optional	Optional
LU	Alignment Up	Required	Optional	LU		Optional				Required				Optional	Optional
MB	Missing Brick	Required	Optional	MB		Optional					Required	Optional		Optional	Optional
MCU	Camera Underwater	Required	Optional	MCU		Optional								Optional	Optional
MGO	General Observation	Required	Optional	MGO										Optional	Required
MGP	General Photo	Required	Optional	MGP										Optional	Optional
MJL	Joint Length Change	Required	Optional	MJL				Required						Optional	Optional
MLC	Lining Change	Required	Optional	MLC										Optional	Optional
MMC	Material Change	Required	Optional	MMC										Optional	Required
MML	Mortar Missing Large	Required	Optional	MM		Optional	L					Required	Optional	Optional	Optional
MMM	Mortar Missing Medium	Required	Optional	MM		Optional	M					Required	Optional	Optional	Optional
MMS	Mortar Missing Small	Required	Optional	MM		Optional	S					Required	Optional	Optional	Optional
MSA	Abandoned Survey	Required	Optional	MSA										Optional	Required
MSC	Shape or Size Change	Required	Optional	MSC				Required	Optional					Optional	Optional
MWL	Water Level	Required	Optional	MWL						Required				Optional	Optional
MWLS	Water Level Sag	Required	Optional	MWL	S	Optional				Required				Optional	Optional
MWM	Water Mark	Required	Optional	MWM						Required				Optional	Optional
MYN	Dye Test Not Visible	Required	Optional	MY	N									Optional	Optional
MYV	Dye Test Visible	Required	Optional	MY	V									Optional	Required
OB	Obstacle Brick	Required	Optional	OB		Optional				Required		Required	Optional	Optional	Optional
OBC	Obstacle Thru Connection	Required	Optional	OBC		Optional				Required		Required	Optional	Optional	Optional
OBI	Obstacle Intruding Thru Wall	Required	Optional	OBI		Optional				Required		Required	Optional	Optional	Optional
OBJ	Obstacle In Joint	Required	Optional	OBJ		Optional				Required	Required	Required	Optional	Optional	Optional
OBM	Obstacle Pipe Material	Required	Optional	OBM		Optional				Required		Required	Optional	Optional	Optional
OB	Obstacle Construction Debris	Required	Optional	OB		Optional				Required		Required	Optional	Optional	Optional
OBP	Obstacle External Pipe or Cable	Required	Optional	OBP		Optional				Required		Required	Optional	Optional	Optional
OBR	Obstacle Rocks	Required	Optional	OBR		Optional				Required		Required	Optional	Optional	Optional
OBS	Obstacle Built Into Structure	Required	Optional	OBS		Optional				Required		Required	Optional	Optional	Optional
OBZ	Obstacle Other	Required	Optional	OBZ		Optional				Required		Required	Optional	Optional	Required
RBB	Roots Ball Barrel	Required	Optional	RB	B	Optional				Required > 50%		Required	Optional	Optional	Optional
RBC	Roots Ball Connection	Required	Optional	RB	C	Optional				Required > 50%		Required	Optional	Optional	Optional
RBJ	Roots Ball Joint	Required	Optional	RBJ		Optional				Required > 50%	Required	Required	Optional	Optional	Optional
RBL	Roots Ball Lateral	Required	Optional	RB	L	Optional				Required > 50%		Required	Optional	Optional	Optional
RFB	Roots Fine Barrel	Required	Optional	RF	B	Optional						Required	Optional	Optional	Optional
RFC	Roots Fine Connection	Required	Optional	RF	C	Optional						Required	Optional	Optional	Optional
RFJ	Roots Fine Joint	Required	Optional	RFJ		Optional					Required	Required	Optional	Optional	Optional
RFL	Roots Fine Lateral	Required	Optional	RF	L	Optional						Required	Optional	Optional	Optional
RMB	Roots Medium Barrel	Required	Optional	RM	B	Optional				Required ≤ 50%		Required	Optional	Optional	Optional
RMC	Roots Medium Connection	Required	Optional	RM	C	Optional				Required ≤ 50%		Required	Optional	Optional	Optional
RMJ	Roots Medium Joint	Required	Optional	RMJ		Optional				Required ≤ 50%	Required	Required	Optional	Optional	Optional
RML	Roots Medium Lateral	Required	Optional	RM	L	Optional				Required ≤ 50%		Required	Optional	Optional	Optional
RPL	Repair Localized Liner	Required	Optional	RPL		Optional					Optional			Optional	Required
RPLD	Repair Localized Liner Defective	Required	Optional	RPL	D	Optional					Optional			Optional	Required
RPP	Repair Patch	Required	Optional	RPP		Optional					Optional	Required	Optional	Optional	Required
RPPD	Repair Patch Defective	Required	Optional	RPP	D	Optional					Optional	Required	Optional	Optional	Required
RPR	Repair Point Pipe Replaced	Required	Optional	RPR		Optional					Optional			Optional	Required
RPRD	Repair Point Defective	Required	Optional	RPR	D	Optional					Optional			Optional	Required

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				Group/Descriptor	Modifier/Severity		S/M/L	Inches		%		At/ From	To		
								1 st	2 nd						
RPZ	Repair Other	Required	Optional	RPZ		Optional					Optional			Optional	Required
RPZD	Repair Other Defective	Required	Optional	RPZ	D	Optional					Optional			Optional	Required
RTB	Roots Tap Barrel	Required	Optional	RT	B	Optional				Required >1/2" thick		Required	Optional	Optional	Optional
RTC	Roots Tap Connection	Required	Optional	RT	C	Optional				Required >1/2" thick		Required	Optional	Optional	#NAME?
RTJ	Roots Tap Joint	Required	Optional	RTJ		Optional				Required >1/2" thick	Required	Required	Optional	Optional	Optional
RTL	Roots Tap Lateral	Required	Optional	RT	L	Optional				Required >1/2" thick		Required	Optional	Optional	Optional
SAM	Surface Aggregate Missing	Required	Optional	SAM		Optional					Optional	Required	Optional	Optional	Optional
SAMC	Surface Aggregate Missing Chemical	Required	Optional	SAM	C	Optional					Optional	Required	Optional	Optional	Optional
SAMM	Surface Aggregate Missing Mechanical	Required	Optional	SAM	M	Optional					Optional	Required	Optional	Optional	Optional
SAMZ	Surface Aggregate Missing Unknown	Required	Optional	SAM	Z	Optional					Optional	Required	Optional	Optional	Optional
SAP	Surface Aggregate Projecting	Required	Optional	SAP		Optional					Optional	Required	Optional	Optional	Optional
SAPC	Surface Aggregate Projecting Chemical	Required	Optional	SAP	C	Optional					Optional	Required	Optional	Optional	Optional
SAPM	Surface Aggregate Projecting Mechanical	Required	Optional	SAP	M	Optional					Optional	Required	Optional	Optional	Optional
SAPZ	Surface Aggregate Projecting Unknown	Required	Optional	SAP	Z	Optional					Optional	Required	Optional	Optional	Optional
SAV	Surface Aggregate Visible	Required	Optional	SAV		Optional					Optional	Required	Optional	Optional	Optional
SAVC	Surface Aggregate Visible Chemical	Required	Optional	SAV	C	Optional					Optional	Required	Optional	Optional	Optional
SAVM	Surface Aggregate Visible Mechanical	Required	Optional	SAV	M	Optional					Optional	Required	Optional	Optional	Optional
SAVZ	Surface Aggregate Visible Unknown	Required	Optional	SAV	Z	Optional					Optional	Required	Optional	Optional	Optional
SCP	Surface Corrosion Metal Pipe	Required	Optional	SCP		Optional					Optional	Required	Optional	Optional	Optional
SMW	Surface Missing Wall	Required	Optional	SMW		Optional					Optional	Required	Optional	Optional	Optional
SMWC	Surface Missing Wall Chemical	Required	Optional	SMW	C	Optional					Optional	Required	Optional	Optional	Optional
SMWM	Surface Missing Wall Mechanical	Required	Optional	SMW	M	Optional					Optional	Required	Optional	Optional	Optional
SMWZ	Surface Missing Wall Unknown	Required	Optional	SMW	Z	Optional					Optional	Required	Optional	Optional	Optional
SRC	Surface Reinforcement Corroded	Required	Optional	SRC		Optional					Optional	Required	Optional	Optional	Optional
SRCC	Surface Reinforcement Corroded Chemical	Required	Optional	SRC	C	Optional					Optional	Required	Optional	Optional	Optional
SRCM	Surface Reinforcement Corroded Mechanical	Required	Optional	SRC	M	Optional					Optional	Required	Optional	Optional	Optional
SRCZ	Surface Reinforcement Corroded Unknown	Required	Optional	SRC	Z	Optional					Optional	Required	Optional	Optional	Optional
SRI	Surface Roughness Increased	Required	Optional	SRI		Optional					Optional	Required	Optional	Optional	Optional
SRIC	Surface Roughness Increased Chemical	Required	Optional	SRI	C	Optional					Optional	Required	Optional	Optional	Optional
SRIM	Surface Roughness Increased Mechanical	Required	Optional	SRI	M	Optional					Optional	Required	Optional	Optional	Optional
SRIZ	Surface Roughness Increased Unknown	Required	Optional	SRI	Z	Optional					Optional	Required	Optional	Optional	Optional
SRP	Surface Reinforcement Projecting	Required	Optional	SRP		Optional					Optional	Required	Optional	Optional	Optional

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				Group/Descriptor	Modifier/Severity		S/M/L	Inches		%		At/ From	To		
								1 st	2 nd						
SRPC	Surface Reinforcement Projecting Chemical	Required	Optional	SRP	C	Optional					Optional	Required	Optional	Optional	Optional
SRPM	Surface Reinforcement Projecting Mechanical	Required	Optional	SRP	M	Optional					Optional	Required	Optional	Optional	Optional
SRPZ	Surface Reinforcement Projecting Unknown	Required	Optional	SRP	Z	Optional					Optional	Required	Optional	Optional	Optional
SRV	Surface Reinforcement Visible	Required	Optional	SRV		Optional					Optional	Required	Optional	Optional	Optional
SRVC	Surface Reinforcement Visible Chemical	Required	Optional	SRV	C	Optional					Optional	Required	Optional	Optional	Optional
SRVM	Surface Reinforcement Visible Mechanical	Required	Optional	SRV	M	Optional					Optional	Required	Optional	Optional	Optional
SRVZ	Surface Reinforcement Visible Unknown	Required	Optional	SRV	Z	Optional					Optional	Required	Optional	Optional	Optional
SSS	Surface Spalling	Required	Optional	SSS		Optional					Optional	Required	Optional	Optional	Optional
SSSC	Surface Spalling Chemical	Required	Optional	SSS	C	Optional					Optional	Required	Optional	Optional	Optional
SSSM	Surface Spalling Mechanical	Required	Optional	SSS	M	Optional					Optional	Required	Optional	Optional	Optional
SSSZ	Surface Spalling Other	Required	Optional	SSS	Z	Optional					Optional	Required	Optional	Optional	Optional
SZ	Surface Other	Required	Optional	SZ		Optional					Optional	Required	Optional	Optional	Required
SZC	Surface Other Chemical	Required	Optional	SZ	C	Optional					Optional	Required	Optional	Optional	Required
SZM	Surface Other Mechanical	Required	Optional	SZ	M	Optional					Optional	Required	Optional	Optional	Required
SZZ	Surface Other Unknown	Required	Optional	SZ	Z	Optional					Optional	Required	Optional	Optional	Required
TB	Tap Break-in	Required	Optional	TB				Required			Optional	Required		Optional	Optional
TBA	Tap Break-in Active	Required	Optional	TB	A			Required			Optional	Required		Optional	Optional
TBB	Tap Break-in Abandoned	Required	Optional	TB	B			Required			Optional	Required		Optional	Optional
TBC	Tap Break-in Capped	Required	Optional	TB	C			Required			Optional	Required		Optional	Optional
TBD	Tap Break-in Defective	Required	Optional	TB	D			Required	Optional		Optional	Required		Optional	Optional
TBI	Tap Break-in Intruding	Required	Optional	TB	I			Required	Required		Optional	Required		Optional	Optional
TF	Tap Factory	Required	Optional	TF				Required			Optional	Required		Optional	Optional
TFA	Tap Factory Active	Required	Optional	TF	A			Required			Optional	Required		Optional	Optional
TFB	Tap Factory Abandoned	Required	Optional	TF	B			Required			Optional	Required		Optional	Optional
TFC	Tap Factory Capped	Required	Optional	TF	C			Required			Optional	Required		Optional	Optional
TFD	Tap Factory Defective	Required	Optional	TF	D			Required	Optional		Optional	Required		Optional	Optional
TFI	Tap Factory Intruding	Required	Optional	TF	I			Required	Required		Optional	Required		Optional	Optional
TR	Tap Rehabilitated	Required	Optional	TR				Required			Optional	Required		Optional	Optional
TRD	Tap Rehabilitated Defective	Required	Optional	TR	D			Required	Optional		Optional	Required		Optional	Optional
TRI	Tap Rehabilitated Intruding	Required	Optional	TR	I			Required	Required		Optional	Required		Optional	Optional
TS	Tap Saddle	Required	Optional	TS				Required			Optional	Required		Optional	Optional
TSA	Tap Saddle Active	Required	Optional	TS	A			Required			Optional	Required		Optional	Optional
TSB	Tap Saddle Abandoned	Required	Optional	TS	B			Required			Optional	Required		Optional	Optional
TSC	Tap Saddle Capped	Required	Optional	TS	C			Required			Optional	Required		Optional	Optional
TSD	Tap Saddle Defective	Required	Optional	TS	D			Required	Optional		Optional	Required		Optional	Optional
TSI	Tap Saddle Intruding	Required	Optional	TS	I			Required	Required		Optional	Required		Optional	Optional
VC	Vermin Cockroach	Required	Optional	VC							Optional			Optional	Optional
VR	Vermin Rat	Required	Optional	VR				Required			Optional			Optional	Optional
VZ	Vermin Other	Required	Optional	VZ				Required			Optional			Optional	Optional
WFC	Weld Failure Circumferential	Required	Optional	WFC		Optional					Optional	Required	Optional	Optional	Optional
WFL	Weld Failure Longitudinal	Required	Optional	WFL		Optional					Optional	Required	Optional	Optional	Optional
WFM	Weld Failure Multiple	Required	Optional	WFM		Optional					Optional	Required	Optional	Optional	Optional
WFS	Weld Failure Spiral	Required	Optional	WFS		Optional					Optional	Required	Optional	Optional	Optional

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				Group/Descriptor	Modifier/Severity		S/M/L	Inches		%		At/ From	To		
								1 st	2 nd						
WFZ	Weld Failure Other	Required	Optional	WFZ		Optional				Optional	Required	Optional	Optional	Required	
XB (brick pipe only)	Collapse Brick Sewer	Required	Optional	XB					Required > 40%				Optional	Optional	
XP	Collapse Pipe Sewer	Required	Optional	XP					Required > 40%				Optional	Optional	

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Appendix B - Family Group and Group Order

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Family	Group	Full PACP Code	Description	Distance (Feet)	Video Ref.	Code		Continuous	S/M/L	Value			Joint	Location		Image Ref.	Remarks
						Group/Descriptor	Modifier/Severity			Inches		%		At/ From	To		
										1 st	2 nd						
Structural	Broken	B	Broken	Required	Optional	B		Optional		Optional	Optional	Optional	Required	Optional	Optional	Optional	
Structural	Broken	BSV	Broken Soil Visible	Required	Optional	B	SV	Optional		Optional	Optional	Optional	Required	Optional	Optional	Optional	
Structural	Broken	BVV	Broken Void Visible	Required	Optional	B	VV	Optional		Optional	Optional	Optional	Required	Optional	Optional	Optional	
Structural	Crack	CC	Crack Circumferential	Required	Optional	CC		Optional		Optional	Optional	Optional	Required	Required	Optional	Optional	
Structural	Crack	CH2	Crack Longitudinal Hinge, 2	Required	Optional	CH2		Optional		Optional	Optional	Optional	Required	Required	Optional	Optional	
Structural	Crack	CH3	Crack Longitudinal Hinge, 3	Required	Optional	CH3		Optional		Optional	Optional	Optional	Required	Required	Optional	Optional	
Structural	Crack	CH4	Crack Longitudinal Hinge, 4	Required	Optional	CH4		Optional		Optional	Optional	Optional	Required	Required	Optional	Optional	
Structural	Crack	CL	Crack Longitudinal	Required	Optional	CL		Optional		Optional	Optional	Optional	Required		Optional	Optional	
Structural	Crack	CM	Crack Multiple	Required	Optional	CM		Optional		Optional	Optional	Optional	Required	Required	Optional	Optional	
Structural	Crack	CS	Crack Spiral	Required	Optional	CS		Optional		Optional	Optional	Optional	Required	Required	Optional	Optional	
Structural	Deformed	D	Deformed	Required	Optional	D		Optional				Required 5% increments			Optional	Optional	
Structural	Brickwork	DB	Displaced Brick	Required	Optional	DB		Optional					Required	Optional	Optional	Optional	
Structural	Deformed	DH	Deformed Horizontal	Required	Optional	DH		Optional				Required 5% increments			Optional	Optional	
Structural	Brickwork	DI	Dropped Invert	Required	Optional	DI		Optional		Required					Optional	Optional	
Structural	Deformed	DV	Deformed Vertical	Required	Optional	DV		Optional				Required 5% increments			Optional	Optional	
Structural	Fracture	FC	Fracture Circumferential	Required	Optional	FC		Optional		Optional	Optional	Optional	Required	Required	Optional	Optional	
Structural	Fracture	FH2	Fracture Longitudinal Hinge, 2	Required	Optional	FH2		Optional		Optional	Optional	Optional	Required	Required	Optional	Optional	
Structural	Fracture	FH3	Fracture Longitudinal Hinge, 3	Required	Optional	FH3		Optional		Optional	Optional	Optional	Required	Required	Optional	Optional	
Structural	Fracture	FH4	Fracture Longitudinal Hinge, 4	Required	Optional	FH4		Optional		Optional	Optional	Optional	Required	Required	Optional	Optional	
Structural	Fracture	FL	Fracture Longitudinal	Required	Optional	FL		Optional		Optional	Optional	Optional	Required		Optional	Optional	
Structural	Fracture	FM	Fracture Multiple	Required	Optional	FM		Optional		Optional	Optional	Optional	Required	Required	Optional	Optional	
Structural	Fracture	FS	Fracture Spiral	Required	Optional	FS		Optional		Optional	Optional	Optional	Required	Required	Optional	Optional	
Structural	Hole	H	Hole	Required	Optional	H		Optional		Optional	Optional	Optional	Required	Optional	Optional	Optional	
Structural	Hole	HSV	Hole Soil Visible	Required	Optional	H	SV	Optional		Optional	Optional	Optional	Required	Optional	Optional	Optional	
Structural	Hole	HWV	Hole Void Visible	Required	Optional	H	VV	Optional		Optional	Optional	Optional	Required	Optional	Optional	Optional	
Structural	Joint	JAL	Joint Angular Large	Required	Optional	JA		Optional	L	Optional	Optional				Optional	Optional	
Structural	Joint	JAM	Joint Angular Medium	Required	Optional	JA		Optional	M	Optional	Optional				Optional	Optional	
Structural	Joint	JOL	Joint Offset Large	Required	Optional	JO		Optional	L	Optional	Optional				Optional	Optional	
Structural	Joint	JOM	Joint Offset Medium	Required	Optional	JO		Optional	M	Optional	Optional				Optional	Optional	
Structural	Joint	JSL	Joint Separated Large	Required	Optional	JS		Optional	L	Optional	Optional				Optional	Optional	
Structural	Joint	JSM	Joint Separated Medium	Required	Optional	JS		Optional	M	Optional	Optional				Optional	Optional	
Structural	Buckling	KW	Buckling Wall	Required	Optional	K	W	Optional					Required	Optional	Optional	Optional	
Structural	Buckling	KD	Buckling Dimpling	Required	Optional	K	D	Optional					Required	Optional	Optional	Optional	
Structural	Buckling	KI	Inverse Curvature	Required	Optional	K	I	Optional				required		Required	Optional	Optional	
Structural	Lining	LFAC	Lining Failure Abandoned Connection	Required	Optional	LF	AC	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFAS	Lining Failure Annular Space	Required	Optional	LF	AS	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFB	Lining Failure Blistered	Required	Optional	LF	B	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFBK	Lining Failure Buckled	Required	Optional	LF	BK	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFBU	Lining Failure Bulged	Required	Optional	LF	BU	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFCS	Lining Failure Connection Cut Shifted	Required	Optional	LF	CS	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFD	Lining Failure Detached	Required	Optional	LF	D	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFDC	Lining Failure Discoloration	Required	Optional	LF	DC	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFDE	Lining Failure Defective End	Required	Optional	LF	DE	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFDL	Lining Failure Delaminating	Required	Optional	LF	DL	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFOC	Lining Failure Overcut Connection	Required	Optional	LF	OC	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFPH	Lining Failure Pinhole	Required	Optional	LF	PH	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFPS	Lining Failure Resin slug	Required	Optional	LF	PS	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFUC	Lining Failure Undercut Connection	Required	Optional	LF	UC	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFW	Lining Failure Wrinkled	Required	Optional	LF	W	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Lining	LFZ	Lining Failure Other	Required	Optional	LF	Z	Optional				Optional	Required	Optional	Optional	Required	
Structural	Brickwork	MB	Missing Brick	Required	Optional	MB		Optional					Required	Optional	Optional	Optional	

Appendix B - Family Group and Group Order

Family	Group	Full PACP Code	Description	Distance (Feet)	Video Ref.	Code		Continuous	S/M/L	Value			Joint	Location		Image Ref.	Remarks
						Group/Descriptor	Modifier/Severity			1"	2"	%		At/ From	To		
Structural	Brickwork	MML	Mortar Missing Large	Required	Optional	MM		Optional	L					Required	Optional	Optional	Optional
Structural	Brickwork	MMM	Mortar Missing Medium	Required	Optional	MM		Optional	M					Required	Optional	Optional	Optional
Structural	Brickwork	MMS	Mortar Missing Small	Required	Optional	MM		Optional	S					Required	Optional	Optional	Optional
Structural	Point Repair	RPL	Repair Localized Liner	Required	Optional	RPL		Optional					Optional			Optional	Required
Structural	Point Repair	RPLD	Repair Localized Liner Defective	Required	Optional	RPL	D	Optional					Optional			Optional	Required
Structural	Point Repair	RPP	Repair Patch	Required	Optional	RPP		Optional					Optional	Required	Optional	Optional	Required
Structural	Point Repair	RPPD	Repair Patch Defective	Required	Optional	RPP	D	Optional					Optional	Required	Optional	Optional	Required
Structural	Point Repair	RPR	Repair Point Pipe Replaced	Required	Optional	RPR		Optional					Optional			Optional	Required
Structural	Point Repair	RPRD	Repair Point Defective	Required	Optional	RPR	D	Optional					Optional			Optional	Required
Structural	Point Repair	RPZ	Repair Other	Required	Optional	RPZ		Optional					Optional			Optional	Required
Structural	Point Repair	RPZD	Repair Other Defective	Required	Optional	RPZ	D	Optional					Optional			Optional	Required
Structural	Surface Damage	SAM	Surface Aggregate Missing	Required	Optional	SAM		Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SAMC	Surface Aggregate Missing Chemical	Required	Optional	SAM	C	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SAMM	Surface Aggregate Missing Mechanical	Required	Optional	SAM	M	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SAMZ	Surface Aggregate Missing Unknown	Required	Optional	SAM	Z	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SAP	Surface Aggregate Projecting	Required	Optional	SAP		Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SAPC	Surface Aggregate Projecting Chemical	Required	Optional	SAP	C	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SAPM	Surface Aggregate Projecting Mechanical	Required	Optional	SAP	M	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SAPZ	Surface Aggregate Projecting Unknown	Required	Optional	SAP	Z	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SAV	Surface Aggregate Visible	Required	Optional	SAV		Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SAVC	Surface Aggregate Visible Chemical	Required	Optional	SAV	C	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SAVM	Surface Aggregate Visible Mechanical	Required	Optional	SAV	M	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SAVZ	Surface Aggregate Visible Unknown	Required	Optional	SAV	Z	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SCP	Surface Corrosion Metal Pipe	Required	Optional	SCP		Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SMW	Surface Missing Wall	Required	Optional	SMW		Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SMWC	Surface Missing Wall Chemical	Required	Optional	SMW	C	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SMWM	Surface Missing Wall Mechanical	Required	Optional	SMW	M	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SMWZ	Surface Missing Wall Unknown	Required	Optional	SMW	Z	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SRC	Surface Reinforcement Corroded	Required	Optional	SRC		Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SRCC	Surface Reinforcement Corroded Chemical	Required	Optional	SRC	C	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SRCM	Surface Reinforcement Corroded Mechanical	Required	Optional	SRC	M	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SRCZ	Surface Reinforcement Corroded Unknown	Required	Optional	SRC	Z	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SRI	Surface Roughness Increased	Required	Optional	SRI		Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SRIC	Surface Roughness Increased Chemical	Required	Optional	SRI	C	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SRIM	Surface Roughness Increased Mechanical	Required	Optional	SRI	M	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SRIZ	Surface Roughness Increased Unknown	Required	Optional	SRI	Z	Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SRP	Surface Reinforcement Projecting	Required	Optional	SRP		Optional					Optional	Required	Optional	Optional	Optional
Structural	Surface Damage	SRPC	Surface Reinforcement Projecting Chemical	Required	Optional	SRP	C	Optional					Optional	Required	Optional	Optional	Optional

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Appendix B - Family Group and Group Order

Family	Group	Full PACP Code	Description	Distance (Feet)	Video Ref.	Code		Continuous	S/M/L	Value			Joint	Location		Image Ref.	Remarks
						Group/Descriptor	Modifier/Severity			1"	2"	%		At/ From	To		
Structural	Surface Damage	SRPM	Surface Reinforcement Projecting Mechanical	Required	Optional	SRP	M	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Surface Damage	SRPZ	Surface Reinforcement Projecting Unknown	Required	Optional	SRP	Z	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Surface Damage	SRV	Surface Reinforcement Visible	Required	Optional	SRV		Optional				Optional	Required	Optional	Optional	Optional	
Structural	Surface Damage	SRVC	Surface Reinforcement Visible Chemical	Required	Optional	SRV	C	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Surface Damage	SRVM	Surface Reinforcement Visible Mechanical	Required	Optional	SRV	M	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Surface Damage	SRVZ	Surface Reinforcement Visible Unknown	Required	Optional	SRV	Z	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Surface Damage	SSS	Surface Spalling	Required	Optional	SSS		Optional				Optional	Required	Optional	Optional	Optional	
Structural	Surface Damage	SSSC	Surface Spalling Chemical	Required	Optional	SSS	C	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Surface Damage	SSSM	Surface Spalling Mechanical	Required	Optional	SSS	M	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Surface Damage	SSSZ	Surface Spalling Other	Required	Optional	SSS	Z	Optional				Optional	Required	Optional	Optional	Optional	
Structural	Surface Damage	SC	Surface Other	Required	Optional	SC		Optional				Optional	Required	Optional	Optional	Required	
Structural	Surface Damage	SZC	Surface Other Chemical	Required	Optional	SZ	C	Optional				Optional	Required	Optional	Optional	Required	
Structural	Surface Damage	SZM	Surface Other Mechanical	Required	Optional	SZ	M	Optional				Optional	Required	Optional	Optional	Required	
Structural	Surface Damage	SZZ	Surface Other Unknown	Required	Optional	SZ	Z	Optional				Optional	Required	Optional	Optional	Required	
Structural	Weld Failure	WFC	Weld Failure Circumferential	Required	Optional	WFC		Optional				Optional	Required	Optional	Optional	Optional	
Structural	Weld Failure	WFL	Weld Failure Longitudinal	Required	Optional	WFL		Optional				Optional	Required	Optional	Optional	Optional	
Structural	Weld Failure	WFM	Weld Failure Multiple	Required	Optional	WFM		Optional				Optional	Required	Optional	Optional	Optional	
Structural	Weld Failure	WFS	Weld Failure Spiral	Required	Optional	WFS		Optional				Optional	Required	Optional	Optional	Optional	
Structural	Weld Failure	WFZ	Weld Failure Other	Required	Optional	WFZ		Optional				Optional	Required	Optional	Optional	Required	
Structural	Collapse	XB (brick pipe only)	Collapse Brick Sewer	Required	Optional	XB					Required > 40%				Optional	Optional	
Structural	Collapse	XP	Collapse Pipe Sewer	Required	Optional	XP					Required > 40%				Optional	Optional	
Operational	Deposits	DAE	Deposits Attached Encrustation	Required	Optional	DA	E	Optional			Required	Optional	Required	Optional	Optional	Optional	
Operational	Deposits	DAGS	Deposits Attached Grease	Required	Optional	DA	GS	Optional			Required	Optional	Required	Optional	Optional	Optional	
Operational	Deposits	DAR	Deposits Attached Ragging	Required	Optional	DA	R	Optional			Required	Optional	Required	Optional	Optional	Optional	
Operational	Deposits	DAZ	Deposits Attached Other	Required	Optional	DA	Z	Optional			Required	Optional	Required	Optional	Optional	Optional	
Operational	Deposits	DNF	Deposits Ingressed Fine	Required	Optional	DN	F	Optional			Required	Optional	Required	Optional	Optional	Optional	
Operational	Deposits	DNGV	Deposits Ingressed Gravel	Required	Optional	DN	GV	Optional			Required	Optional	Required	Optional	Optional	Optional	
Operational	Deposits	DNZ	Deposits Ingressed Other	Required	Optional	DN	Z	Optional			Required	Optional	Required	Optional	Optional	Required	
Operational	Deposits	DSC	Deposits Settled Compacted	Required	Optional	DS	C	Optional			Required	Optional	Required	Optional	Optional	Optional	
Operational	Deposits	DSF	Deposits Settled Fine	Required	Optional	DS	F	Optional			Required	Optional	Required	Optional	Optional	Optional	
Operational	Deposits	DSGV	Deposits Settled Gravel	Required	Optional	DS	GV	Optional			Required	Optional	Required	Optional	Optional	Optional	
Operational	Deposits	DSZ	Deposits Settled Other	Required	Optional	DS	Z	Optional			Required	Optional	Required	Optional	Optional	Required	
Operational	Grout Test	GRT	Grout done at Location	Required	Optional					Optional					Optional	Optional	
Operational	Grout Test	GTFJ	Grout Air Test Fail Joint	Required	Optional					Optional	Required				Optional	Optional	
Operational	Grout Test	GTFL	Grout Air Test Fail Lateral	Required	Optional					Optional	Required				Optional	Optional	
Operational	Grout Test	GTPJ	Grout Air Test Pass Joint	Required	Optional										Optional	Optional	
Operational	Grout Test	GTPL	Grout Air Test Pass Lateral	Required	Optional										Optional	Optional	
Operational	Grout Test	GTUJ	Grout Air Test Unable Joint	Required	Optional										Optional	Required	
Operational	Grout Test	GTUL	Grout Air Test Unable Lateral	Required	Optional										Optional	Required	
Operational	Infil	ID	Infil Dropper	Required	Optional	ID		Optional				Optional	Required	Optional	Optional	Optional	
Operational	Infil	IG	Infil Gusher	Required	Optional	IG		Optional				Optional	Required	Optional	Optional	Optional	
Operational	Infil	IR	Infil Runner	Required	Optional	IR		Optional				Optional	Required	Optional	Optional	Optional	
Operational	Infil	IS	Infil Stain	Required	Optional	IS		Optional				Optional	Required	Optional	Optional	Optional	
Operational	Infil	IW	Infil Weeper	Required	Optional	IW		Optional				Optional	Required	Optional	Optional	Optional	
Operational	Obstacles	OBB	Obstacle Brick	Required	Optional	OBB		Optional			Required		Required	Optional	Optional	Optional	
Operational	Obstacles	OBC	Obstacle Thru Connection	Required	Optional	OBC		Optional			Required		Required	Optional	Optional	Optional	
Operational	Obstacles	OBI	Obstacle Intruding Thru Wall	Required	Optional	OBI		Optional			Required		Required	Optional	Optional	Optional	
Operational	Obstacles	OBJ	Obstacle In Joint	Required	Optional	OBJ		Optional			Required	Required	Required	Optional	Optional	Optional	
Operational	Obstacles	OBN	Obstacle Pipe Material	Required	Optional	OBN		Optional			Required		Required	Optional	Optional	Optional	
Operational	Obstacles	OBN	Obstacle Construction Debris	Required	Optional	OBN		Optional			Required		Required	Optional	Optional	Optional	

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Appendix B - Family Group and Group Order

Family	Group	Full PACP Code	Description	Distance (Feet)	Video Ref.	Code		Continuous	S/M/L	Value			Joint	Location		Image Ref.	Remarks
						Group/Descriptor	Modifier/Severity			1"	2"	%		At/ From	To		
Operational	Obstacles	OBP	Obstacle External Pipe or Cable	Required	Optional	OBP		Optional				Required		Required	Optional	Optional	Optional
Operational	Obstacles	OBR	Obstacle Rocks	Required	Optional	OBR		Optional				Required		Required	Optional	Optional	Optional
Operational	Obstacles	OBS	Obstacle Built Into Structure	Required	Optional	OBS		Optional				Required		Required	Optional	Optional	Optional
Operational	Obstacles	OBZ	Obstacle Other	Required	Optional	OBZ		Optional				Required		Required	Optional	Optional	Required
Operational	Roots	RBB	Roots Ball Barrel	Required	Optional	RB	B	Optional				Required > 50%		Required	Optional	Optional	Optional
Operational	Roots	RBC	Roots Ball Connection	Required	Optional	RB	C	Optional				Required > 50%		Required	Optional	Optional	Optional
Operational	Roots	RBJ	Roots Ball Joint	Required	Optional	RBJ		Optional				Required > 50%	Required	Required	Optional	Optional	Optional
Operational	Roots	RBL	Roots Ball Lateral	Required	Optional	RB	L	Optional				Required > 50%		Required	Optional	Optional	Optional
Operational	Roots	RFB	Roots Fine Barrel	Required	Optional	RF	B	Optional						Required	Optional	Optional	Optional
Operational	Roots	RFC	Roots Fine Connection	Required	Optional	RF	C	Optional						Required	Optional	Optional	Optional
Operational	Roots	RFJ	Roots Fine Joint	Required	Optional	RFJ		Optional					Required	Required	Optional	Optional	Optional
Operational	Roots	RFL	Roots Fine Lateral	Required	Optional	RF	L	Optional						Required	Optional	Optional	Optional
Operational	Roots	RMB	Roots Medium Barrel	Required	Optional	RM	B	Optional				Required < 50%		Required	Optional	Optional	Optional
Operational	Roots	RMC	Roots Medium Connection	Required	Optional	RM	C	Optional				Required < 50%		Required	Optional	Optional	Optional
Operational	Roots	RMJ	Roots Medium Joint	Required	Optional	RMJ		Optional				Required < 50%	Required	Required	Optional	Optional	Optional
Operational	Roots	RMI	Roots Medium Lateral	Required	Optional	RM	L	Optional				Required < 50%		Required	Optional	Optional	Optional
Operational	Roots	RTB	Roots Tap Barrel	Required	Optional	RT	B	Optional				Required > 1/2" thick		Required	Optional	Optional	Optional
Operational	Roots	RTC	Roots Tap Connection	Required	Optional	RT	C	Optional				Required > 1/2" thick		Required	Optional	Optional	#NAME?
Operational	Roots	RTJ	Roots Tap Joint	Required	Optional	RTJ		Optional				Required > 1/2" thick	Required	Required	Optional	Optional	Optional
Operational	Roots	RTL	Roots Tap Lateral	Required	Optional	RT	L	Optional				Required > 1/2" thick		Required	Optional	Optional	Optional
Operational	Vermin	VC	Vermin Cockroach	Required	Optional	VC							Optional			Optional	Optional
Operational	Vermin	VR	Vermin Rat	Required	Optional	VR				Required			Optional			Optional	Optional
Operational	Vermin	VZ	Vermin Other	Required	Optional	VZ				Required			Optional			Optional	Optional
Construction	Line	LL	Alignment Left	Required	Optional	LL		Optional				Required				Optional	Optional
Construction	Line	LLD	Alignment Left Down	Required	Optional	LLD		Optional				Required				Optional	Optional
Construction	Line	LLU	Alignment Left Up	Required	Optional	LLU		Optional				Required				Optional	Optional
Construction	Line	LR	Alignment Right	Required	Optional	LR		Optional				Required				Optional	Optional
Construction	Line	LRD	Alignment Right Down	Required	Optional	LRD		Optional				Required				Optional	Optional
Construction	Line	LRU	Alignment Right Up	Required	Optional	LRU		Optional				Required				Optional	Optional
Construction	Line	LU	Alignment Up	Required	Optional	LU		Optional				Required				Optional	Optional
Construction	Access Point	ACB	Catch Basin	Required	Optional	ACB										Optional	Optional
Construction	Access Point	ACOH	Cleanout House	Required	Optional	ACO	H									Optional	Optional
Construction	Access Point	ACOM	Cleanout Mainline	Required	Optional	ACO	M									Optional	Required
Construction	Access Point	ACOP	Cleanout Propertyline	Required	Optional	ACO	P									Optional	Optional
Construction	Access Point	ADP	Discharge Point	Required	Optional	ADP										Optional	Optional
Construction	Access Point	AEP	End of Pipe	Required	Optional	AEP										Optional	Optional
Construction	Access Point	AJB	Junction Box	Required	Optional	AJB										Optional	Required
Construction	Access Point	AM	Meter	Required	Optional	AM										Optional	Required
Construction	Access Point	AMH	Manhole	Required	Optional	AMH										Optional	Required
Construction	Access Point	AOC	Special Chamber	Required	Optional	AOC										Optional	Optional
Construction	Access Point	ATC	Tree Connection	Required	Optional	ATC			Required	Optional			Required		Optional	Optional	Optional
Construction	Access Point	AWA	Wastewater Access Device	Required	Optional	AWA										Optional	Required
Construction	Access Point	AWW	Wet Well	Required	Optional	AWW										Optional	Optional
Construction	Intruding Seal	ISGT	Intruding Sealing GROUT	Required	Optional	ISGT		Optional				Required		Required	Required	Optional	Optional
Construction	Intruding Seal	ISSR	Intruding Sealing Ring	Required	Optional	ISSR		Optional				Required		Required	Required	Optional	Optional
Construction	Intruding Seal	ISSRB	Intruding Sealing Ring Broken	Required	Optional	ISSR	B	Optional				Required		Required	Required	Optional	Optional
Construction	Intruding Seal	ISSRH	Intruding Sealing Ring Hanging	Required	Optional	ISSR	H	Optional				Required		Required	Required	Optional	Optional
Construction	Intruding Seal	ISSRL	Intruding Sealing Ring Loose/Poorly Fitting	Required	Optional	ISSR	L	Optional				Required		Required	Required	Optional	Optional
Construction	Intruding Seal	ISZ	Intruding Sealing Other	Required	Optional	ISZ		Optional				Required		Required	Required	Optional	Optional
Construction	Alignment	LD	Alignment Down	Required	Optional	LD		Optional				Required				Optional	Optional
Construction	Tap	TB	Tap Break-in	Required	Optional	TB				Required			Optional	Required		Optional	Optional
Construction	Tap	TBA	Tap Break-in Active	Required	Optional	TB	A			Required			Optional	Required		Optional	Optional
Construction	Tap	TBB	Tap Break-in Abandoned	Required	Optional	TB	B			Required			Optional	Required		Optional	Optional
Construction	Tap	TBC	Tap Break-in Cased	Required	Optional	TB	C			Required			Optional	Required		Optional	Optional
Construction	Tap	TBD	Tap Break-in Defective	Required	Optional	TB	D			Required	Optional		Optional	Required		Optional	Optional
Construction	Tap	TBI	Tap Break-in Intruding	Required	Optional	TB	I			Required	Required		Optional	Required		Optional	Optional
Construction	Tap	TF	Tap Factory	Required	Optional	TF				Required			Optional	Required		Optional	Optional
Construction	Tap	TFA	Tap Factory Active	Required	Optional	TF	A			Required			Optional	Required		Optional	Optional
Construction	Tap	TFB	Tap Factory Abandoned	Required	Optional	TF	B			Required			Optional	Required		Optional	Optional

Appendix B - Family Group and Group Order

Family	Group	Full PACP Code	Description	Distance (Feet)	Video Ref.	Code		Continuous	S/M/L	Value			Joint	Location		Image Ref.	Remarks
						Group/Descriptor	Modifier/Severity			1 st	2 nd	%		At/ From	To		
Construction	Tap	TFC	Tap Factory Capped	Required	Optional	TF	C			Required			Optional	Required		Optional	Optional
Construction	Tap	TFD	Tap Factory Defective	Required	Optional	TF	D			Required	Optional		Optional	Required		Optional	Optional
Construction	Tap	TFI	Tap Factory Intruding	Required	Optional	TF	I			Required	Required		Optional	Required		Optional	Optional
Construction	Tap	TR	Tap Rehabilitated	Required	Optional	TR				Required			Optional	Required		Optional	Optional
Construction	Tap	TRD	Tap Rehabilitated Defective	Required	Optional	TR	D			Required	Optional		Optional	Required		Optional	Optional
Construction	Tap	TRI	Tap Rehabilitated Intruding	Required	Optional	TR	I			Required	Required		Optional	Required		Optional	Optional
Construction	Tap	TS	Tap Saddle	Required	Optional	TS				Required			Optional	Required		Optional	Optional
Construction	Tap	TSA	Tap Saddle Active	Required	Optional	TS	A			Required			Optional	Required		Optional	Optional
Construction	Tap	TSB	Tap Saddle Abandoned	Required	Optional	TS	B			Required			Optional	Required		Optional	Optional
Construction	Tap	TSC	Tap Saddle Capped	Required	Optional	TS	C			Required			Optional	Required		Optional	Optional
Construction	Tap	TSD	Tap Saddle Defective	Required	Optional	TS	D			Required	Optional		Optional	Required		Optional	Optional
Construction	Tap	TSI	Tap Saddle Intruding	Required	Optional	TS	I			Required	Required		Optional	Required		Optional	Optional
Miscellaneous	Misc	MCU	Camera Underwater	Required	Optional	MCU		Optional								Optional	Optional
Miscellaneous	Misc	MGO	General Observation	Required	Optional	MGO										Optional	Required
Miscellaneous	Misc	MGP	General Photo	Required	Optional	MGP										Optional	Optional
Miscellaneous	Misc	MDL	Joint Length Change	Required	Optional	MDL				Required						Optional	Optional
Miscellaneous	Misc	MLC	Lining Change	Required	Optional	MLC										Optional	Optional
Miscellaneous	Misc	MMC	Material Change	Required	Optional	MMC										Optional	Required
Miscellaneous	Misc	MSA	Abandoned Survey	Required	Optional	MSA										Optional	Required
Miscellaneous	Misc	MSC	Shape or Size Change	Required	Optional	MSC				Required	Optional					Optional	Optional
Miscellaneous	Misc	MWL	Water Level	Required	Optional	MWL						Required				Optional	Optional
Miscellaneous	Misc	MWLS	Water Level Sag	Required	Optional	MWL	S	Optional				Required				Optional	Optional
Miscellaneous	Misc	MWM	Water Mark	Required	Optional	MWM						Required				Optional	Optional
Miscellaneous	Misc	MYN	Dye Test Not Visible	Required	Optional	MY	N									Optional	Optional
Miscellaneous	Misc	MYV	Dye Test Visible	Required	Optional	MY	V									Optional	Required

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Section 4—Continuous Defect Coding

"TRULY" 4-1
"Truly" continuous defects run along the sewer without any interruption for more than three feet (1 meter).
Examples:
- Longitudinal Fractures
- Longitudinal Cracks

"REPEATED" 4-1
"Repeated" continuous defects occur at regular intervals along the sewer. These occur at pipe joints and include:
- Encrustation
- Open Joints
- Circumferential Fractures

Code Changes in Version 6.0.1
Added:
Buckling Wall (KW), Buckling Dimpling (KD), and Buckling Inverse Curvature (KI)

Section 5—Structural Defect Coding (Module 6A)

C CRACK 5-1
CL Longitudinal 5-2
CC Circumferential 5-2
CM Multiple 5-2
CS Spiral 5-2
CH Hinge 5-2

F FRACTURE 5-7
FL Longitudinal 5-7
FC Circumferential 5-7
FM Multiple 5-7
FS Spiral 5-7
FH Hinge 5-7

B BROKEN 5-15
BSV -Soil Visible Beyond Defect 5-15
BV V-Void Visible Beyond Defect 5-15

H HOLE 5-17
HSV -Soil Visible Beyond Defect 5-17
HV V-Void Visible Beyond Defect 5-17

D DEFORMED 5-19
DV Deformed Vertically (brick) 5-19
DH Deformed Horizontally (brick) 5-19

X COLLAPSE 5-23
XP Pipe Collapse 5-23
XB Brick Collapse 5-23

J JOINT 5-26
JO Joint Offset (Displaced) 5-26
JS Joint Separated (Open) 5-26
JA Joint Angular 5-26

S SURFACE DAMAGE 5-31
SRI Roughness Increased 5-31
SRI - M - Mechanical 5-31
SRI - C - Chemical 5-31
SRI - Z - Not Evident 5-31

S SURFACE DAMAGE 5-31
SAV Aggregate Visible 5-31
SAV - M - Mechanical 5-31
SAV - C - Chemical 5-31
SAV - Z - Not Evident 5-31

S SURFACE DAMAGE 5-31
SAP Aggregate Projecting 5-31
SAP - M - Mechanical 5-31
SAP - C - Chemical 5-31
SAP - Z - Not Evident 5-31

S SURFACE DAMAGE 5-31
SAM Aggregate Missing 5-31
SAM - M - Mechanical 5-31
SAM - C - Chemical 5-31
SAM - Z - Not Evident 5-31

S SURFACE DAMAGE 5-31
SRV Reinforcement Visible 5-31
SRV - M - Mechanical 5-31
SRV - C - Chemical 5-31
SRV - Z - Not Evident 5-31

S SURFACE DAMAGE 5-31
SRP Reinforcement Projecting 5-31
SRP - M - Mechanical 5-31
SRP - C - Chemical 5-31
SRP - Z - Not Evident 5-31

S SURFACE DAMAGE 5-31
SRC Reinforcement Corroded 5-31
SRC - M - Mechanical 5-31
SRC - C - Chemical 5-31
SRC - Z - Not Evident 5-31

S SURFACE DAMAGE 5-31
SNW Missing Wall 5-32
SNW - M - Mechanical 5-32
SNW - C - Chemical 5-32
SNW - Z - Not Evident 5-32

S SURFACE DAMAGE 5-31
SSS Surface Spalling 5-32
SSS - M - Mechanical 5-32
SSS - C - Chemical 5-32
SSS - Z - Not Evident 5-32

S SURFACE DAMAGE 5-31
SZ Other 5-32
SZ - M - Mechanical 5-32
SZ - C - Chemical 5-32
SZ - Z - Not Evident 5-32

S SURFACE DAMAGE 5-31
SCP Corrosion (metal pipe) *no modifiers used 5-32

K BUCKLING 5-45
KW Wall 5-45
KD Dimpling 5-45
KI Inverse Curvature 5-45

LF LINING FAILURE 5-49
LFD Detached Lining 5-49
LFDL Defective End 5-49
LFB Blistered Lining 5-49
LFCS Service Cut Shifted 5-49
LFAC Abandoned Connection 5-49

LF LINING FAILURE 5-49
LFAC Abandoned Connection 5-49
LFCS Service Cut Shifted 5-49
LFB Blistered Lining 5-49
LFD Detached Lining 5-49
LFDL Defective End 5-49
LFUC Undercut Service 5-49
LFBK Buckled Lining 5-49
LFW Wrinkled Lining 5-49
LFAS Annular Space 5-49

LF LINING FAILURE 5-50
LFBU Bulges 5-50
LFDL Discoloration 5-50
LFDL Delamination 5-50
LFPS Resin Slag 5-50
LUPH Pinholes 5-50
LFZ Other 5-50

WF WELD FAILURE 5-67
WFL Longitudinal 5-67
WFC Circumferential 5-67
WFM Multiple 5-67
WFS Spiral 5-67
WFZ Unidentified 5-67

RP POINT REPAIR 5-71
RPR Pipe Replaced 5-69
RPR - D - Defective 5-69
RPP Patch Repair 5-69
RPP - D - Defective 5-69

RP POINT REPAIR 5-71
RPL Localized Pipeliner 5-69
RPL - D - Defective 5-69
RPZ Other 5-69
RPZ - D - Defective 5-69

BRICKWORK 5-77
DB Displaced 5-75
MB Missing 5-75
DI Dropped Invert 5-75

BRICKWORK 5-77
MM Missing Mortar 5-75
S -Small 5-75
M -Medium 5-75
L -Large 5-75

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Section 6—Operational and Maintenance (Module 6B)

D DEPOSITS 6-1	D DEPOSITS 6-1	D DEPOSITS 6-1	R ROOTS 6-7	R ROOTS 6-7	R ROOTS 6-7
DA Attached 6-1	DS Settled 6-1	DN (continued) 6-1	RF Fine 6-7	RT Tap 6-7	RB Ball 6-7
DAE -Encrustation 6-2	DSF -Fine 6-2	DNF -Fine 6-3	RFB -Barrel 6-7	RTB -Barrel 6-7	RBB -Barrel 6-7
DAGS -Grease 6-2	DSGV -Gravel 6-2	DNGV -Gravel (silt & sand) 6-3	RFL -Lateral 6-7	RTL -Lateral 6-7	RBL -Lateral 6-7
DAR -Ragging 6-2	DSC -Hard/Compacted 6-2	DNGV -Gravel 6-3	RFC -Connection 6-8	RTC -Connection 6-8	RBC -Connection 6-8
DAZ -Other 6-2	DSZ -Other 6-2	DNZ -Other 6-3			
I INFILTRATION 6-13	OB OBSTACLES/ OBSTRUCTIONS 6-19	OB OBSTACLES/ OBSTRUCTIONS 6-19	V VERMIN 6-31	G GROUT TEST & SEAL 6-33	G GROUT TEST & SEAL 6-33
IS Stain 6-13	OBB Brick or 6-19	OBS Built into 6-20	VR Rat 6-31	GTP Grout Test 6-33	GTU Grout Test 6-33
IW Weeper 6-13	OBM Pipe Material 6-19	OBN Construction structure 6-20	VC Cockroach 6-31	GTP - J - Joint 6-33	GTU - J - Joint 6-33
IR Runner 6-13	OBI Object protruding 6-19	OBR Rocks 6-20	VZ Other 6-31	GTF Grout Test 6-33	GTU - L - Lateral 6-33
IG Gusher 6-13	OBI Object protruding 6-19	OBS Other 6-20		GTF Grout Test 6-33	GRT Grout Test 6-33
					Location 6-33

Section 7—Construction Features Coding (Module 6C)

T TAP 7-1	T TAP 7-1	T TAP 7-1	T TAP 7-1	IS INTRUDING SEALING MATERIAL 7-9	IS INTRUDING SEALING MATERIAL 7-9
TF Factory Made 7-1	TB Break In/Hammer 7-2	TS Saddle 7-2	TR Rehabilitated 7-2	ISGT Grout 7-9	ISGT Grout 7-9
TFI -Intruding 7-2	TBI -Intruding 7-2	TSA -Active 7-2	TRI -Intruding 7-2	ISZ Other 7-9	ISZ Other 7-9
TFA -Active 7-2	TBA -Active 7-2	TSC -Capped 7-2	TRA -Active 7-2		
TFC -Capped 7-2	TBC -Capped 7-2	TSD -Defective 7-2	TRC -Capped 7-2		
TTF -Abandoned 7-2	TBB -Abandoned 7-2		TRB -Abandoned 7-2		
TTFD -Defective 7-2	TBD -Defective 7-2		TRD -Defective 7-2		
L LINE 7-11	L LINE 7-11	A ACCESS POINT 7-13	A ACCESS POINT 7-13	A ACCESS POINT 7-13	A ACCESS POINT 7-13
LL (of sewer) 7-11	LRU Right Up 7-11	AMH Manhole 7-13	AOC Other Special 7-13	ACO Clean Out 7-14	ACB Catch Basin 7-14
LLU Left Up 7-11	LRL Right Down 7-11	AWA Wastewater Access 7-13	AM Meter 7-13	ACOM -Mainline 7-14	AEP End of Pipe 7-14
LLD Left Down 7-11	LU Up 7-11	ADP Discharge Point 7-13	AWW Wet Well 7-14	ACOP -Property 7-14	
LLR Right 7-11	LD Down 7-11	ATC Tee Connection 7-13	AJB Junction Box 7-14	ACOH -House 7-14	

Section 8—Miscellaneous Features Coding (Module 6D)

M MISCELLANEOUS FEATURES 8-1	M MISCELLANEOUS FEATURES 8-1	M MISCELLANEOUS FEATURES 8-1
MCU Camera Underwater 8-1	MLC Lining Change 8-2	MWM Water Mark 8-2
MGO General Observation 8-1	MMC Material Change 8-2	MY Dye Test 8-2
MGP General Photograph 8-1	MSA Survey Abandoned 8-2	MYV -Dye Visible 8-3
MSC Shape/Size Change 8-1	MWL Water Level 8-2	MYN -Not Visible 8-3
MJL Joint Length Change 8-1	MWLS -Sag 8-2	

Updated November 2010